Course: Cryptography and Network Security
Code: CSN15403
Branch: B.Tech -5th Semester

Lecture – 12 : ASYMMETRIC-KEY CRYPTOGRAPHY

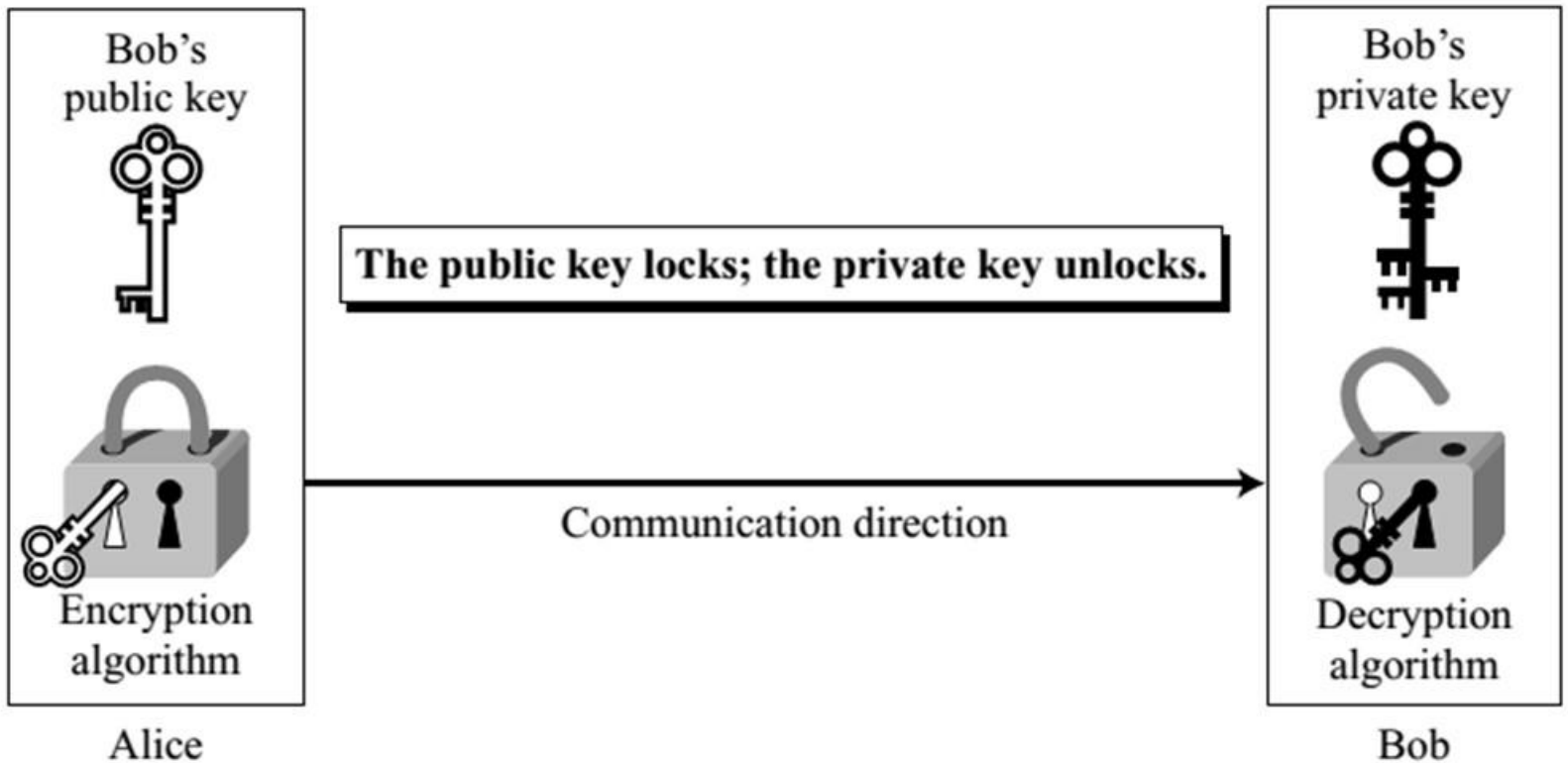
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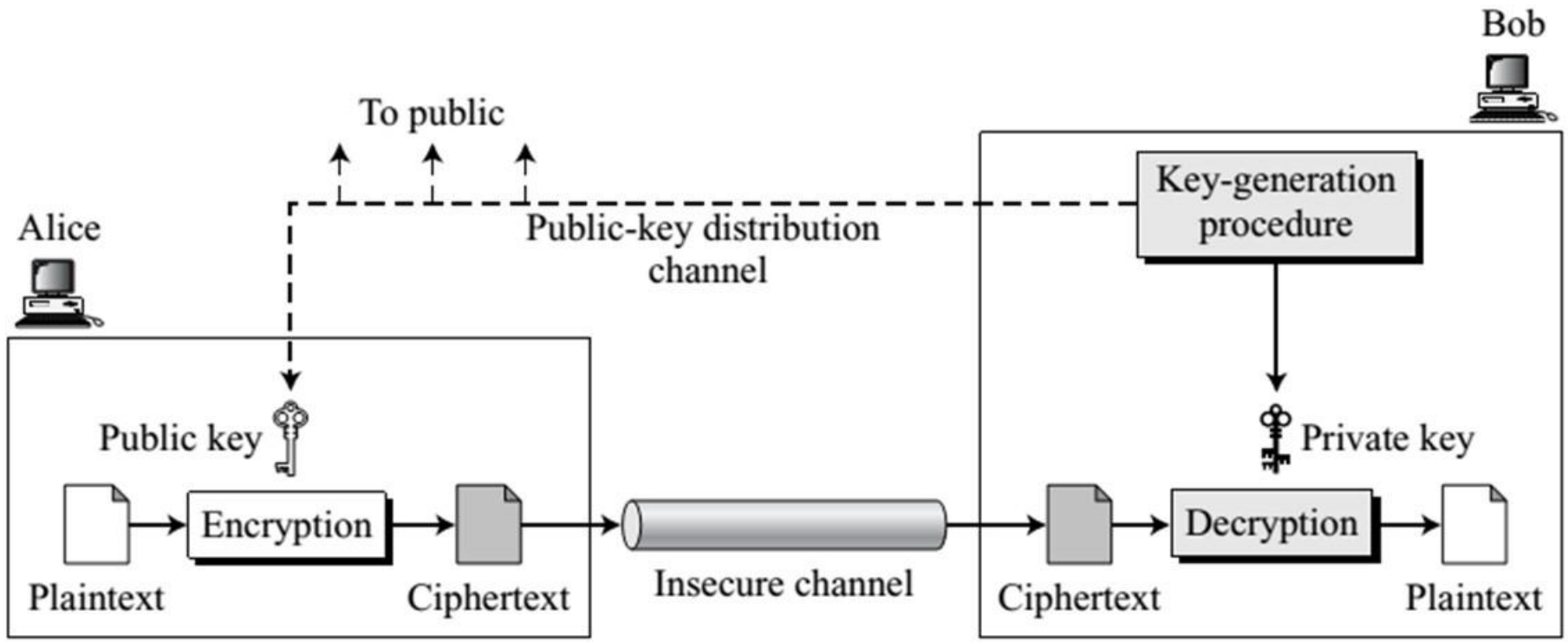
Asymmetric-Key Cryptography

- Locking and unlocking in asymmetric-key cryptosystem



Asymmetric-Key Cryptography

- General idea of asymmetric-key cryptosystem



Asymmetric-Key Cryptography

- Encryption/Decryption
  - The ciphertext can be thought of as $C = f(K_{public}, P)$;
  - The plaintext can be thought of as $P = g(K_{private}, C)$.
  - The function f is used only for encryption;
  - The function g is used only for decryption.
- Need for Both
  - Asymmetric-key cryptography is much slower than symmetric-key cryptography
  - Asymmetric-key cryptography is still needed for authentication, digital signatures, and secret-key exchanges.

Trapdoor One-Way Function

- The main idea behind asymmetric-key cryptography is the concept of the trapdoor oneway function.
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One-Way Function

A **one-way function (OWF)** is a function that satisfies the following two properties:

1. f is easy to compute. In other words, given x, $y = f(x)$ can be easily computed.
2. f^{-1} is difficult to compute. In other words, given y, it is computationally infeasible to calculate $x = f^{-1}(y)$.

Trapdoor One-Way Function

A **trapdoor one-way function (TOWF)** is a one-way function with a third property:

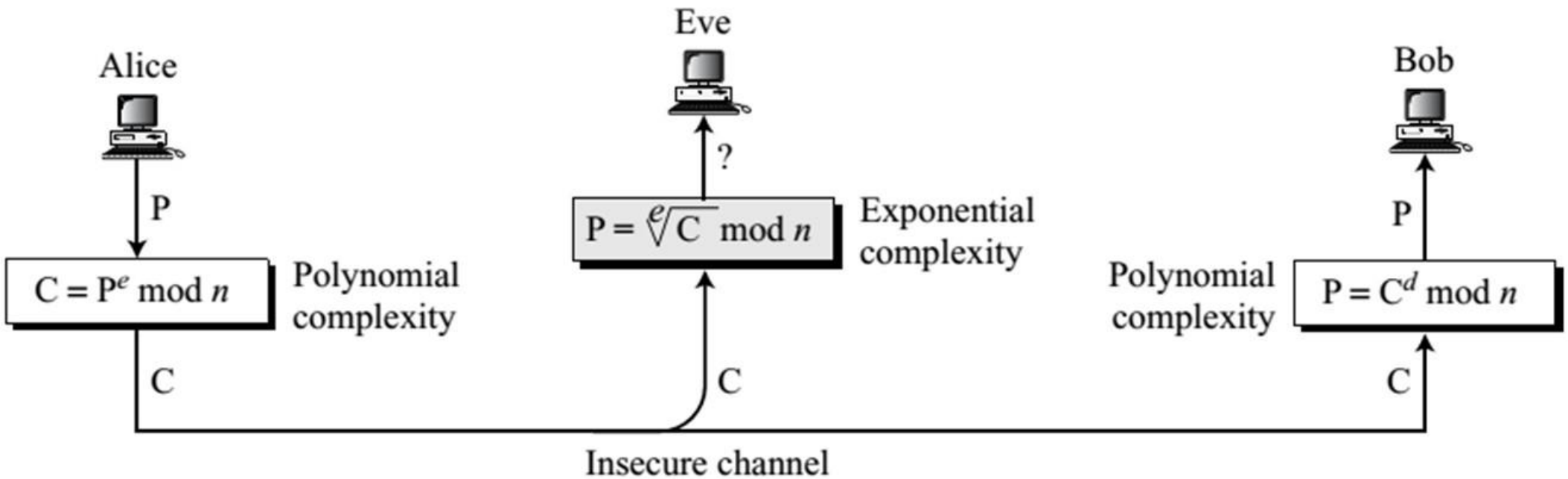
3. Given y and a **trapdoor** (secret), x can be computed easily.

Trapdoor One-Way Function

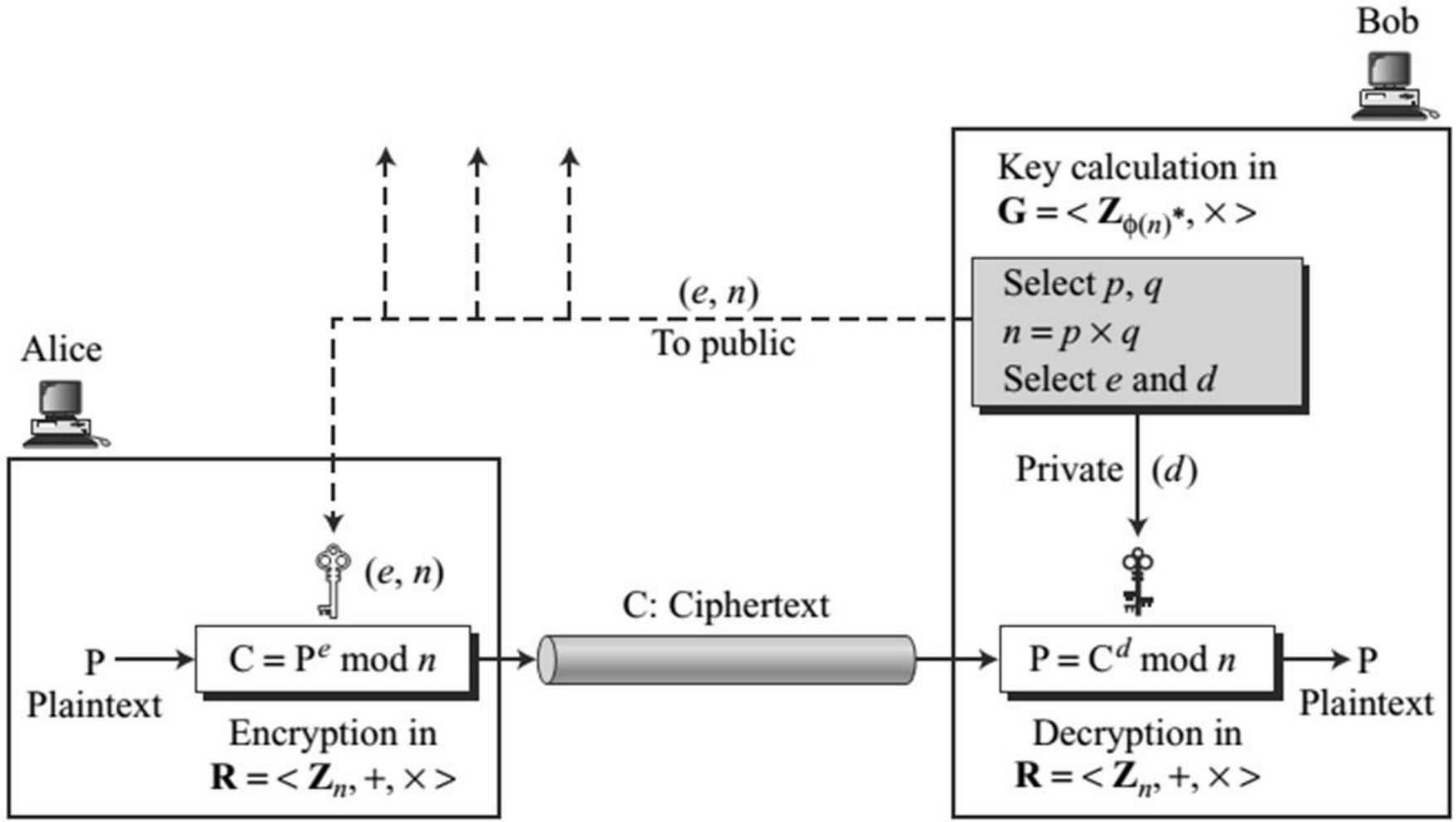
- When n is large, $n = p \times q$ is a one-way function.
- In this function x is a tuple (p, q) of two primes and y is n.
- Given p and q, it is always easy to calculate n; given n, it is very difficult to compute p and q.
- This is the factorization problem.
- There is not a polynomial time solution to the f^{-1} function in this case.
- When n is large, the function $y = x^k \bmod n$ is a trapdoor one-way function.
- Given x, k, and n, it is easy to calculate y using the fast exponential algorithm
- Given y, k, and n, it is very difficult to calculate x.
- However, if we know the trapdoor, k' such that $k \times k' = 1 \bmod \varphi(n)$, we can use $x = y^{k'} \bmod n$ to find x.

RSA CRYPTOSYSTEM

- The most common public-key algorithm is the RSA cryptosystem, named for its inventors (Rivest, Shamir, and Adleman).



Encryption, decryption, and key generation in RSA



RSA Key Generation

RSA_Key_Generation

{

Select two large primes p and q such that $p \neq q$.

$n \leftarrow p \times q$

$\phi(n) \leftarrow (p-1) \times (q-1)$

Select e such that $1 < e < \phi(n)$ and e is coprime to $\phi(n)$

$d \leftarrow e^{-1} \bmod \phi(n)$ // d is inverse of e modulo $\phi(n)$

Public_key $\leftarrow$ (e, n) // To be announced publicly

Private_key $\leftarrow d$ // To be kept secret

return Public_key and Private_key

}

In RSA, the tuple (e, n) is the public key; the integer d is the private key.

Encryption and Decryption

```
RSA_Encryption (P, e, n)                    // P is the plaintext in Z_n and P < n
{
    C ← Fast_Exponentiation (P, e, n)       // Calculation of (P^e mod n)
    return C
}

RSA_Decryption (C, d, n)                    //C is the ciphertext in Z_n
{
    P ← Fast_Exponentiation (C, d, n)       // Calculation of (C^d mod n)
    return P
}
```

Some Trivial Examples

Bob chooses 7 and 11 as p and q and calculates $n = 7 \times 11 = 77$. The value of $\phi(n) = (7 - 1)(11 - 1)$ or 60. Now he chooses two exponents, e and d, from $\mathbf{Z}_{60}$*. If he chooses e to be 13, then d is 37. Note that $e \times d \bmod 60 = 1$ (they are inverses of each other). Now imagine that Alice wants to send the plaintext 5 to Bob. She uses the public exponent 13 to encrypt 5.

Plaintext: 5	$C = 5^{13} = 26 \bmod 77$	Ciphertext: 26

Bob receives the ciphertext 26 and uses the private key 37 to decipher the ciphertext:

Ciphertext: 26	$P = 26^{37} = 5 \bmod 77$	Plaintext: 5

The plaintext 5 sent by Alice is received as plaintext 5 by Bob.

Some Trivial Examples

Now assume that another person, John, wants to send a message to Bob. John can use the same public key announced by Bob (probably on his website), 13; John's plaintext is 63. John calculates the following:

Plaintext: 63	$C = 63^{13} = 28 \bmod 77$	Ciphertext: 28

Bob receives the ciphertext 28 and uses his private key 37 to decipher the ciphertext:

Ciphertext: 28	$P = 28^{37} = 63 \bmod 77$	Plaintext: 63

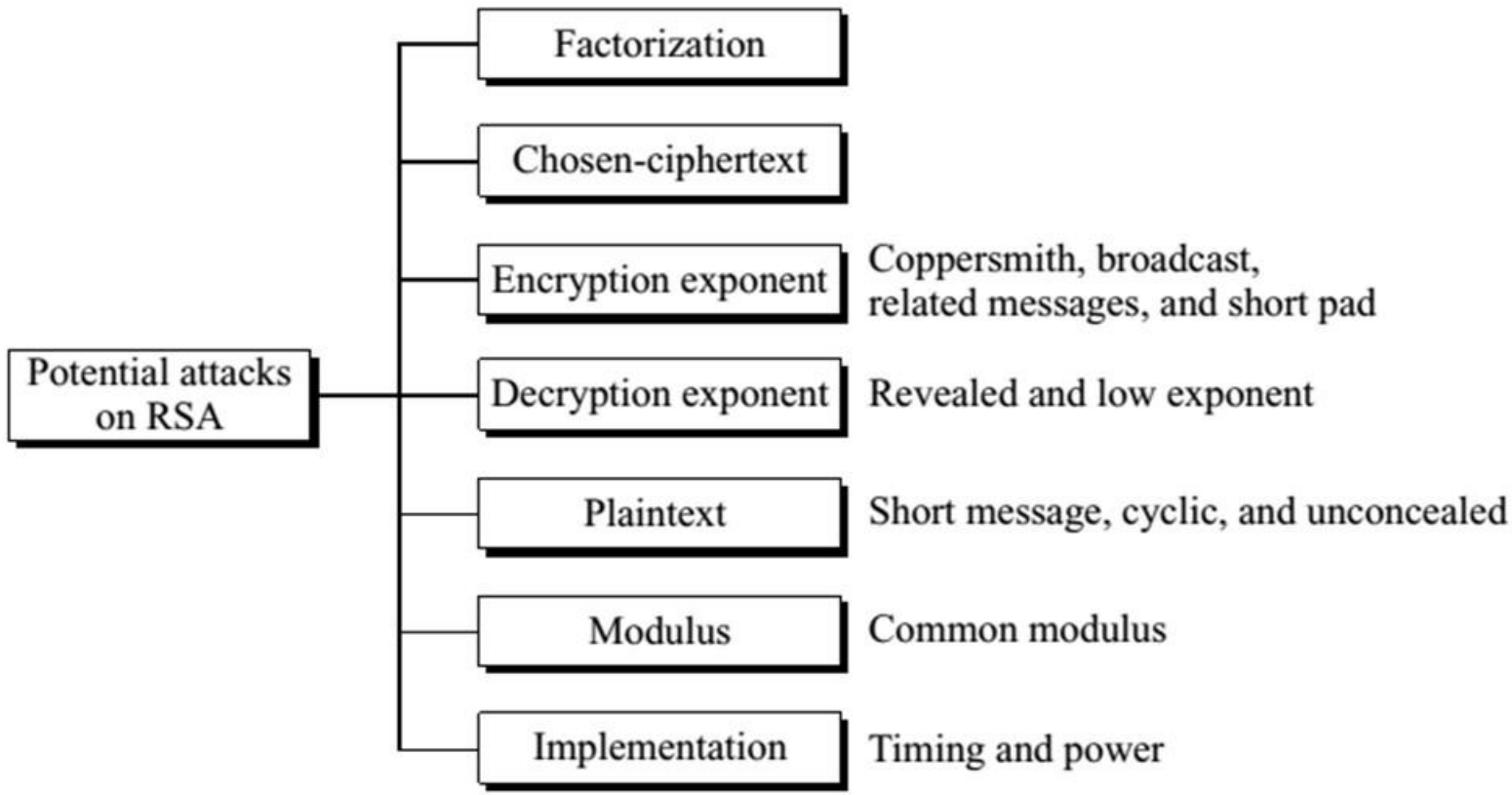
RSA

RSA uses two algebraic structures:
a public ring $R = <Z_n, +, \times>$ and a private group $G = <Z_{\phi(n)}^*, \times>$.

To be secure, the recommended size for each prime, p or q, is 512 bits (almost 154 decimal digits). This makes the size of n, the modulus, 1024 bits (309 digits).

Attacks on RSA

- No devastating attacks on RSA have been yet discovered.



Tutorial

Recommendations

1. The number of bits for n should be at least 1024. This means that n should be around 2^{1024}, or 309 decimal digits.
2. The two primes p and q must each be at least 512 bits. This means that p and q should be around 2^{512} or 154 decimal digits.
3. The values of p and q should not be very close to each other.
4. Both p − 1 and q − 1 should have at least one large prime factor.
5. The ratio p/q should not be close to a rational number with a small numerator or denominator.
6. The modulus n must not be shared.
7. The value of e should be $2^{16} + 1$ or an integer close to this value.
8. If the private key d is leaked, Bob must immediately change n as well as both e and d. It has been proven that knowledge of n and one pair (e, d) can lead to the discovery of other pairs of the same modulus.
9. Messages must be padded using OAEP (Tutorial)